7 - Linear Regression III

Multiple Regression and Ceteris Paribus

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Agenda

Today we ask how multiple regression helps us make more like-for-like comparisons by accounting for other observed differences between laptops.

We will cover:

1. Multiple regression and partial interpretation
2. Controls and ceteris paribus comparisons
3. Categorical variables in linear regression
4. Adjusted R^2 and joint F-test intuition

Recommended reading: Wooldridge, Chapter 3.1–3.3 and Chapter 4.5

1. Why Multiple Regression?

The Problem with a Simple Comparison

Suppose we estimate:

$$Price = \beta_0 + \beta_1 RAM + u$$

The slope compares laptops with different amounts of **RAM**.

But high-ram laptops may also differ in:

- CPU speed;
- brand;
- storage;
- screen size and resolution;
- product segment.

So the simple-regression slope may mix together several differences between laptops.

A More Realistic Pricing Model

A multiple regression model includes more than one explanatory variable:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPU freq + u$$

Interpretation:

- β_1: change in average **price** associated with one more GB of **RAM**, holding **CPUfreq** fixed.
- β_2: change in average **price** associated with one more unit of **CPUfreq**, holding **RAM** fixed.

This is the regression version of a **more comparable comparison**.

2. Interpreting Multiple Regression

The General Multiple Regression Model

With k explanatory variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

The estimated model is:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

For a coefficient $\hat{\beta}_j$:

> *Holding the other **included explanatory variables** fixed, a one-unit increase in x_j is associated with a $\hat{\beta}_j$ unit change in y, on average.*

Multiple Regression in R

```r
m2 <- lm(Price_euros ~ Ram + CPU_freq, data = laptop_reg)
summary(m2)
```

```
Call:
lm(formula = Price_euros ~ Ram + CPU_freq, data = laptop_reg)

Residuals:
     Min       1Q   Median       3Q      Max
-2404.80  -270.60   -80.99   216.63  2780.00

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -229.989     59.808  -3.845 0.000126 ***
Ram           92.590      2.695  34.360  < 2e-16 ***
CPU_freq     253.333     27.265   9.292  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 456.3 on 1272 degrees of freedom
Multiple R-squared:  0.5768,    Adjusted R-squared:  0.5761
F-statistic: 866.7 on 2 and 1272 DF,  p-value: < 2.2e-16
```

Interpretation Practice

For the model:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPU\,freq + u$$

we estimate:

$$\widehat{Price} = -230 + 92.6\,RAM + 253.3\,CPU\,freq$$

Your turn: interpret $\hat{\beta}_1$ and $\hat{\beta}_2$.

Use the words: **associated with**, **on average**, and **holding the other included variable fixed**.

What Controls Can and Cannot Do

Adding controls can help us compare observations that are more similar along the variables we include.

But controls do **not** automatically solve all problems.

Controls help when:

- the variable is relevant;
- it is measured well;
- it belongs in the comparison;
- it captures a real source of confounding.

Controls cannot fix:

- unmeasured factors;
- badly measured variables;
- reverse causality;
- weak research design.

This is why adding controls helps, but does not automatically make a regression causal.

3. Categorical Variables in Linear Regression

Dummy Variables: Comparing Groups

A dummy variable takes values 0 and 1.

For example:

$$Apple = \begin{cases} 1 & \text{if the laptop is Apple} \\ 0 & \text{otherwise} \end{cases}$$

A simple dummy-variable regression is:

$$Price = \beta_0 + \beta_1 Apple + u$$

Then:

- β_0: average price for the reference group, **non-Apple laptops**;
- β_1: average difference between **Apple** and **non-Apple** laptops.

Group Means and Differences Between Group Means

```r
# Mean price by apple status
laptops |>
  mutate(is_apple = ifelse(Company == "Apple", 1, 0)) |>
  group_by(is_apple) |>
  summarise(mean_price = mean(Price_euros, na.rm = TRUE))
```

```
# A tibble: 2 x 2
  is_apple mean_price
     <dbl>      <dbl>
1        0      1128.
2        1      1564.
```

```r
# SLR of Price on is_apple
m3 <- lm(Price_euros ~ is_apple, data = laptop_reg)
summary(m3)$coefficients
```

```
             Estimate Std. Error   t value    Pr(>|t|)
(Intercept) 1127.7810   19.73407 57.148930 0.000000000
is_apple     436.4176  153.76664  2.838181 0.004609169
```

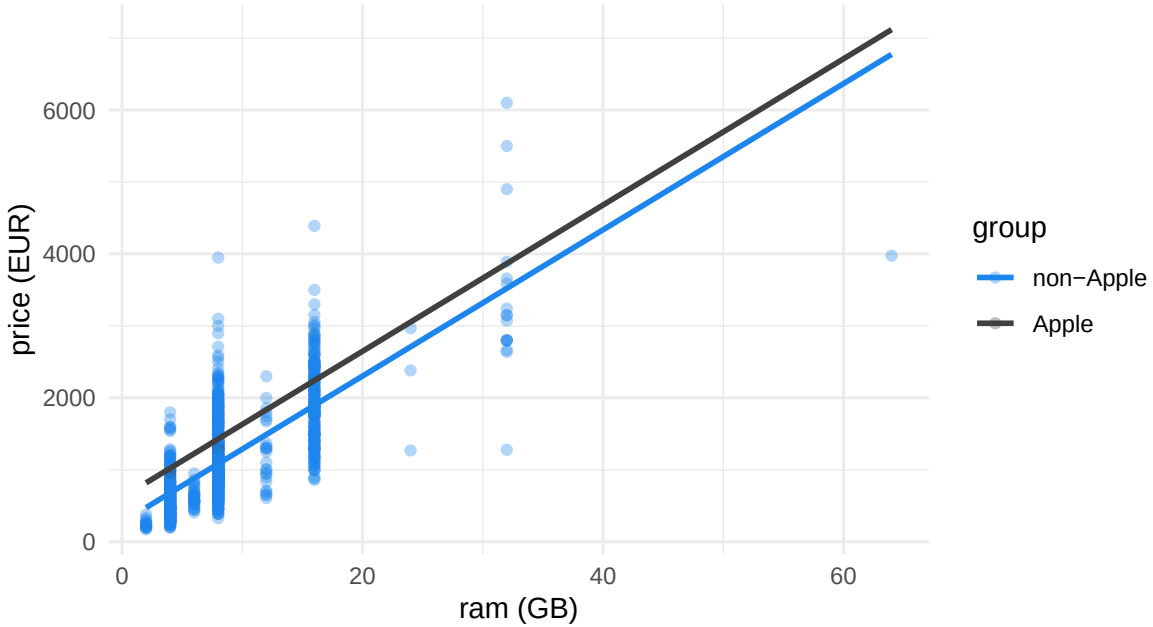
MLR with Dummy Variables

	Price_euros		
	(1)	(2)	(3)
Ram	101.8***		101.6***
	(2.6)		(2.6)
is_apple		436.4***	344.2***
		(153.8)	(103.3)
Constant	276.0***	1,127.8***	272.0***
	(25.5)	(19.7)	(25.5)
Observations	1,275	1,275	1,275
R^2	0.5	0.01	0.6
Adjusted R^2	0.5	0.01	0.6
Residual Std. Error	471.3	698.8	469.4
F Statistic	1,543.5***	8.1***	783.4***
Note:			*p<0.1; **p<0.05; ***p<0.01

When we add *RAM* to the model, β_2 becomes the difference in average price between Apple and non-Apple laptops, holding RAM fixed.

Visualizing a Dummy Control

In this model, $Apple$ shifts the fitted line up or down.



Here the *Apple* coefficient is an **intercept shift** of the line relating *RAM* and *Price*.

More Than Two Categories: Reference Groups

For a categorical variable with m groups, include $m-1$ dummy variables.

Example: operating system with three groups.

os_group	n
Windows	1101
macOS	21
Other	153

If **Windows** is the reference (omitted) group:

- the $MacOS$ coefficient compares macOS laptops with Windows laptops (holding the other variables in the model fixed);
- the $Other$ coefficient compares Other OS laptops with Windows laptops (holding the other variables in the model fixed).

4. Model Fit and Groups of Variables

R^2 and Adjusted R^2

When we add variables, R^2 never decreases if the sample is the same.

That is why R^2 alone is a poor mechanical rule for choosing models.

Adjusted R^2 applies a penalty for adding variables:

$$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$$

Use it as a **diagnostic summary**, not as an automatic model-selection machine.

Testing Groups of Variables: Joint F-Test Intuition

Sometimes we want to test whether a **group** of variables adds explanatory power.

Example:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPUfreq + \beta_3 Apple + \beta_4 Weight + \beta_5 Inches + u$$

A joint null might be:

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

Meaning:

After controlling for RAM and CPUfreq, the added variables do not jointly improve the model.

Restricted vs. Unrestricted Models

The joint F-test asks whether the added variables are **jointly useful**.

Restricted model:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPUfreq + u$$

Unrestricted model:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPUfreq + \beta_3 Apple + \beta_4 Weight + \beta_5 Inches + u$$

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0 \qquad H_a : \text{at least one is different from zero}$$

$$F \approx \frac{\text{improvement in fit from the added variables}}{\text{remaining unexplained variation}}$$

A **large F-statistic** means the added variables reduce unexplained variation substantially relative to the noise that remains.

Restricted vs. Unrestricted Models in R

We can compare the **restricted** and **unrestricted** models directly.

```
m_r  <- lm(Price_euros ~ Ram + CPU_freq, data = laptop_reg)
m_ur <- lm(Price_euros ~ Ram + CPU_freq + is_apple + Weight + Inches,
           data = laptop_reg)
anova(m_r, m_ur)
```

```
Analysis of Variance Table

Model 1: Price_euros ~ Ram + CPU_freq
Model 2: Price_euros ~ Ram + CPU_freq + is_apple + Weight + Inches
  Res.Df       RSS Df Sum of Sq      F    Pr(>F)
1   1272 264785439
2   1269 246524303  3  18261136 31.334 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

If the p-value is small, we reject $H_0 : \beta_3 = \beta_4 = \beta_5 = 0$ and conclude that the added variables are **jointly statistically significant**, conditional on the variables already included.

The Global Significance Test

The joint F-test has an important special case that regression software reports automatically.

For the model:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPU\,freq + \beta_3 Apple + \beta_4 Weight + \beta_5 Inches + u$$

we test:

- $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$
- H_a : at least one slope coefficient is different from zero

Restricted model: $Price = \beta_0 + u$ (None of the explanatory variables contributes.)

This is the global F-test reported automatically with a regression.

Global Significance Test in R

	Price_euros			
	(1)	(2)	(3)	(4)
Ram	101.76***	92.59***	91.93***	94.45***
	(2.59)	(2.69)	(2.68)	(2.80)
CPU_freq		253.33***	264.63***	317.09***
		(27.26)	(27.20)	(27.18)
is_apple			437.07***	311.61***
			(100.17)	(98.68)
Weight				37.76
				(35.00)
Inches				−92.79***
				(15.70)
Constant	276.03***	−229.99***	−257.67***	919.28***
	(25.54)	(59.81)	(59.73)	(185.65)
Observations	1,275	1,275	1,275	1,275
R^2	0.55	0.58	0.58	0.61
Adjusted R^2	0.55	0.58	0.58	0.60
Residual Std. Error	471.29	456.25	453.05	440.76
F Statistic	1,543.52***	866.66***	592.32***	390.27***
Note:				*p<0.1; **p<0.05; ***p<0.01

Comparing Models With Different Numbers of Predictors

R^2: Did overall fit increase?

- Always weakly increases when predictors are added.

Adjusted R^2: Did fit improve enough to compensate for the extra predictors?

- Use to compare model fit while accounting for model size.

Joint F**-test:** Do the added predictors jointly contribute to explaining y

- Use when you want to formally test whether a group of added variables contributes to explaining the outcome.

Important: Neither tells us that the larger model is causally better or correctly specified.

5. Closing

What We Learned Today

Key ideas:

- A multiple-regression coefficient compares observations that have the same values of the other **included** explanatory variables.
- Dummy variables compare groups relative to a **reference group**.
- R^2, adjusted R^2, and joint F-tests summarize fit or statistical contribution, not causal credibility.
- Controls help, but they do not automatically solve the problem of a fair comparison.

Next: What would make this more like-for-like comparison credible enough to support a causal interpretation?

Recommended Reading

- Wooldridge, *Introductory Econometrics: A Modern Approach*, 7th ed., Chapter 3.1–3.2 and Chapters 4.4-4.5
  - Section 3.1: Motivation for Multiple Regression
  - Section 3.2: Mechanics and Interpretation of Ordinary Least Squares
  - Section 4.4: Testing Hypotheses About A Single Linear Combination of the Parameters
  - Section 4.5: Testing Multiple Linear Restrictions: the F Test